Rock, Paper, Scissor (Python)

Status	Done
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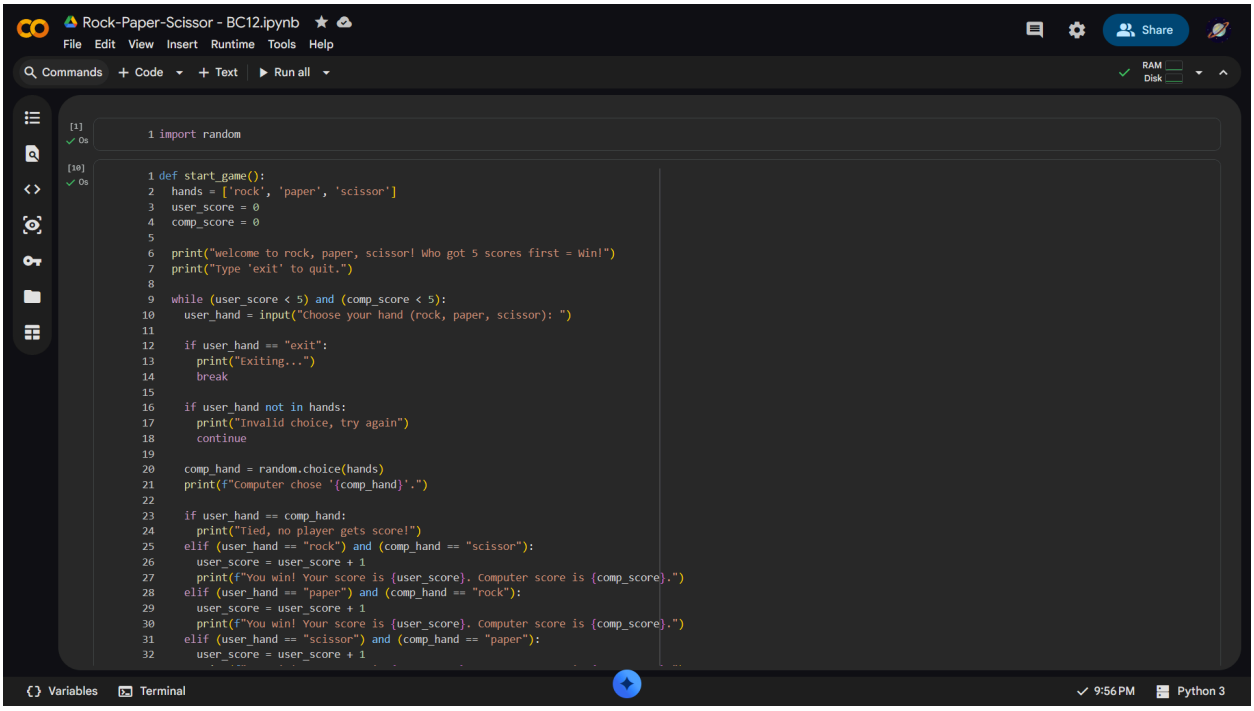
Classes	Live 14 - Intro to Python Programming

This homework is the same with R. Just switches to Python.



My Code

```python
import random

def start_game():
  hands = ['rock', 'paper', 'scissor']
```

```
  user_score = 0
  comp_score = 0

  print("welcome to rock, paper, scissor! Who got 5 scores fi
rst = Win!")
  print("Type 'exit' to quit.")

  while (user_score < 5) and (comp_score < 5):
    user_hand = input("Choose your hand (rock, paper, scisso
r): ")

    if user_hand == "exit":
      print("Exiting...")
      break

    if user_hand not in hands:
      print("Invalid choice, try again")
      continue

    comp_hand = random.choice(hands)
    print(f"Computer chose '{comp_hand}'.")

    if user_hand == comp_hand:
      print("Tied, no player gets score!")
    elif (user_hand == "rock") and (comp_hand == "scissor"):
      user_score = user_score + 1
      print(f"You win! Your score is {user_score}. Computer s
core is {comp_score}.")
    elif (user_hand == "paper") and (comp_hand == "rock"):
      user_score = user_score + 1
      print(f"You win! Your score is {user_score}. Computer s
core is {comp_score}.")
    elif (user_hand == "scissor") and (comp_hand == "paper"):
      user_score = user_score + 1
      print(f"You win! Your score is {user_score}. Computer s
core is {comp_score}.")
```

```
    else:
      comp_score = comp_score + 1
      print(f"Computer wins! Your score is {user_score}. Comp
uter score is {comp_score}")

    if user_score == 5:
      print("Congratulation! You win the round!")
    elif comp_score == 5:
      print("Game Over! Computer wins the round!")
      print("Don't give up, you can try again!")

start_game()
```

To activate, runs `start_game()` function and enjoy!

Details

Let's break down into details.

First is creating function called `start_game()`. This function requires zero argument.

```
def start_game():
```

`hand()` is variable where it stores `rock`, `paper` and `scissor` in list. This is for computer randomly choose data in list.

`user_score` is variable where it stores 0. The purpose is for adding a score for user.

`comp_score` is variable where it stores 0. The purpose is the same except this is for computer.

```
hands = ['rock', 'paper', 'scissor']
user_score = 0
comp_score = 0
```

When starting the game, it shows welcome message.

```
print("welcome to rock, paper, scissor! Who got 5 scores firs
t = Win!")
print("Type 'exit' to quit.")
```

This function/game use `while` loop where it will run if `user_score < 5 AND comp_score < 5`.

And I create `user_hand`, a variable where it receives data by user input.

If user writes something that doesn't match any of `hand` variable, it shows that it's an invalid choice and the function will run `user_hand` again to receive the valid input.

```
while (user_score < 5) and (comp_score < 5):
  user_hand = input("Choose your hand (rock, paper, scissor):
")

  if user_hand == "exit":
    print("Exiting...")
    break

  if user_hand not in hands:
    print("Invalid choice, try again")
    continue
```

And this game allows you to quit anytime by typing `exit`. It will break the loop

We need computer to random the hand. So, I create `comp_hand` variable where it uses `random.choice()` to randomize.

```
comp_hand = random.choice(hands)
```

```
print(f"Computer chose '{comp_hand}'.")
```

This also show what computer chooses.

Now, we have `user_hand` and `comp_hand`. Let's see who will get the score.

From now on, it's literally `if-else` function where I need to see who will win.

Rock, Paper and Scissor scenarios:

- Rock → Scissor = **Win**
- Paper → Rock = **Win**
- Scissor → Paper = **Win**
- If both hands are the same = **Draw**
- Else = **Lose**

We will apply to the code exactly the same as scenario.

```
if user_hand == comp_hand:
  print("Tied, no player gets score!")
elif (user_hand == "rock") and (comp_hand == "scissor"):
  user_score = user_score + 1
  print(f"You win! Your score is {user_score}. Computer score
is {comp_score}.")
elif (user_hand == "paper") and (comp_hand == "rock"):
  user_score = user_score + 1
  print(f"You win! Your score is {user_score}. Computer score
is {comp_score}.")
elif (user_hand == "scissor") and (comp_hand == "paper"):
  user_score = user_score + 1
  print(f"You win! Your score is {user_score}. Computer score
is {comp_score}.")
else:
```

```
  comp_score = comp_score + 1
  print(f"Computer wins! Your score is {user_score}. Computer
score is {comp_score}")
```

If the logic is true (which means user wins) then user gets a score.

Score will add +1 to `user_score`.

If the logic is false (which means user didn't match any logics) then computer gets a score.

Score will add +1 to `comp_score`.

If both hands are the same, no score for both player.

This game, who has reaches 5 score first will win the round.

So, I add this logic.

```
if user_score == 5:
  print("Congratulation! You win the round!")
elif comp_score == 5:
  print("Game Over! Computer wins the round!")
  print("Don't give up, you can try again!")
```

If `user_score == 5`, user wins.

But if `comp_score == 5`, user loses.

And here's the code detail. Let's take a look at gameplay.

Gameplay

Start a game

```
1 start_game()
```

```
welcome to rock, paper, scissor! Who got 5 scores first = Win!
Type 'exit' to quit.
Choose your hand (rock, paper, scissor):
```

User gets a score

```
1 start_game()
```

```
welcome to rock, paper, scissor! Who got 5 scores first = Win!
Type 'exit' to quit.
Choose your hand (rock, paper, scissor): paper
Computer chose 'rock'.
You win! Your score is 1. Computer score is 0.
Choose your hand (rock, paper, scissor):
```

Computer gets a score

```
1 start_game()
```

```
welcome to rock, paper, scissor! Who got 5 scores first = Win!
Type 'exit' to quit.
Choose your hand (rock, paper, scissor): rock
Computer chose 'paper'.
Computer wins! Your score is 0. Computer score is 1
Choose your hand (rock, paper, scissor):
```

Tied

```
1 start_game()
```

```
welcome to rock, paper, scissor! Who got 5 scores first = Win!
Type 'exit' to quit.
Choose your hand (rock, paper, scissor): scissor
Computer chose 'scissor'.
Tied, no player gets score!
Choose your hand (rock, paper, scissor):
```

User wins

```python
start_game()
```

```
welcome to rock, paper, scissor! Who got 5 scores first = Win!
Type 'exit' to quit.
Choose your hand (rock, paper, scissor): scissor
Computer chose 'scissor'.
Tied, no player gets score!
Choose your hand (rock, paper, scissor): scissor
Computer chose 'paper'.
You win! Your score is 1. Computer score is 0.
Choose your hand (rock, paper, scissor): scissor
Computer chose 'paper'.
You win! Your score is 2. Computer score is 0.
Choose your hand (rock, paper, scissor): scissor
Computer chose 'paper'.
You win! Your score is 3. Computer score is 0.
Choose your hand (rock, paper, scissor): scissor
Computer chose 'paper'.
You win! Your score is 4. Computer score is 0.
Choose your hand (rock, paper, scissor): scissor
Computer chose 'rock'.
Computer wins! Your score is 4. Computer score is 1
Choose your hand (rock, paper, scissor): scissor
Computer chose 'scissor'.
Tied, no player gets score!
Choose your hand (rock, paper, scissor): scissor
Computer chose 'paper'.
You win! Your score is 5. Computer score is 1.
Congratulation! You win the round!
```

Computer wins

```
start_game()
```

```
welcome to rock, paper, scissor! Who got 5 scores first = Win!
Type 'exit' to quit.
Choose your hand (rock, paper, scissor): rock
Computer chose 'paper'.
Computer wins! Your score is 0. Computer score is 1
Choose your hand (rock, paper, scissor): rock
Computer chose 'scissor'.
You win! Your score is 1. Computer score is 1.
Choose your hand (rock, paper, scissor): rock
Computer chose 'paper'.
Computer wins! Your score is 1. Computer score is 2
Choose your hand (rock, paper, scissor): rock
Computer chose 'scissor'.
You win! Your score is 2. Computer score is 2.
Choose your hand (rock, paper, scissor): rock
Computer chose 'scissor'.
You win! Your score is 3. Computer score is 2.
Choose your hand (rock, paper, scissor): rock
Computer chose 'paper'.
Computer wins! Your score is 3. Computer score is 3
Choose your hand (rock, paper, scissor): rock
Computer chose 'paper'.
Computer wins! Your score is 3. Computer score is 4
Choose your hand (rock, paper, scissor): rock
Computer chose 'paper'.
Computer wins! Your score is 3. Computer score is 5
Game Over! Computer wins the round!
Don't give up, you can try again!
```

Exit a game

```
start_game()
```

```
welcome to rock, paper, scissor! Who got 5 scores first = Win!
Type 'exit' to quit.
Choose your hand (rock, paper, scissor): exit
Exiting...
```